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Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# **NutriSmart AI: Personalized Nutrition, Recipe Recommendation, and Budget Meal Planning**

**A CAPSTONE PROJECT REPORT**

*Submitted in the partial fulfilment for the Course o*

**CSA4305-Internet Programming**

*to the award of the degree of*

**BACHELOR OF ENGINEERING**

**IN**

**(CSE AI,IT,CSE)**

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## DECLARATION

We, of the **[CSE AI, AI&DS&IT]**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled '**[AI-Based NutriSmart System for Personalized Nutrition, Recipe Recommendation, and Budget Meal Planning]**' is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

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## BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**NutriSmart AI: Personalized Nutrition, Recipe Recommendation, and Budget Meal Planning**” has been carried out by **[Student Names]** under the supervision of **Dr.D.Sudhagar** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech **[CSE -AI,IT,CSE]** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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Signature With Student Name

# Abstract

NutriSmart AI is an intelligent nutrition and meal-planning system designed to address the growing challenges of unhealthy eating habits, limited nutritional awareness, expensive meal choices, and the difficulty of preparing personalized diets. Many existing food and nutrition applications provide generic recipes or calorie information but do not effectively combine individual nutritional requirements, food preferences, available ingredients, and budget constraints into a single recommendation system. NutriSmart AI aims to overcome these limitations by providing personalized nutrition recommendations, healthy recipe suggestions, and budget-conscious meal plans.

The proposed system collects user information such as age, gender, dietary preferences, activity level, nutritional goals, allergies, available ingredients, and daily or weekly food budget. Artificial intelligence and recommendation techniques are then used to estimate nutritional requirements and recommend suitable meals. The system can rank recipes according to nutritional value, user preferences, preparation requirements, ingredient availability, and estimated cost. A meal-planning module generates daily or weekly meal schedules while attempting to maintain an appropriate balance of calories, carbohydrates, proteins, fats, vitamins, and minerals within the specified budget.

The expected outcome is an accessible AI-powered platform that helps users make healthier and more economical food decisions. NutriSmart AI demonstrates how artificial intelligence, data analysis, recommendation systems, and optimization techniques can be integrated to support personalized nutrition and practical meal planning.

**Keywords:** Artificial Intelligence, Personalized Nutrition, Recipe Recommendation, Meal Planning, Machine

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# Chapter 1: Introduction

## 1.1 Background Information

Food plays a fundamental role in maintaining physical health, energy levels, productivity, and overall well-being. However, choosing appropriate food on a daily basis can be difficult because nutritional requirements vary significantly between individuals. Factors such as age, gender, body characteristics, physical activity, dietary preferences, allergies, health objectives, and lifestyle influence the type and quantity of food an individual should consume.

Traditional meal planning often requires users to manually calculate calories, search for suitable recipes, compare ingredient prices, and prepare shopping lists. This process can be time-consuming and difficult for users without nutritional knowledge. In addition, healthy meals are sometimes perceived as expensive, while budget restrictions may lead users to select less nutritious alternatives.

The development of Artificial Intelligence and machine learning provides an opportunity to improve this situation. AI-based recommendation systems can analyze user information and food datasets to identify meals that are more relevant to individual requirements.

NutriSmart AI is proposed as an integrated intelligent platform that combines **personalized nutrition, recipe recommendation, and budget meal planning**. Instead of providing the same recommendations to every user, the system considers individual preferences and constraints.

The platform is designed to answer practical questions such as:

- What should I eat today?
- How many calories do I approximately need?
- Which recipes match my dietary preference?
- What can I cook using ingredients I already have?
- Can I prepare healthy meals within my budget?
- What should my weekly meal plan look like?

## **1.2 Project Objectives**

The main objective of NutriSmart AI is to develop an intelligent system that provides personalized, nutritious, and affordable meal recommendations.

### **Specific objectives**

1. To develop a personalized user nutrition profile.
2. To estimate daily nutritional and calorie requirements based on user inputs.
3. To recommend recipes according to nutritional requirements and dietary preferences.
4. To consider allergies and food restrictions during recommendation.
5. To provide budget-friendly meal plans.
6. To utilize available ingredients when generating recipe recommendations.
7. To create daily and weekly meal schedules.
8. To provide estimated nutritional information for recommended meals.
9. To provide estimated meal and ingredient costs.
10. To develop a simple and user-friendly web interface.
11. To demonstrate the practical application of AI and recommendation techniques in nutrition planning.

## **1.3 Significance of the Project**

NutriSmart AI is significant because it addresses multiple problems simultaneously rather than focusing only on calorie tracking.

The system can benefit students, working professionals, families, fitness-oriented users, and individuals interested in improving their eating habits.

The major contributions include:



### **Personalized nutrition**

Instead of generic food suggestions, the system generates recommendations based on individual user profiles.

### **Affordable healthy eating**

The budget-planning component attempts to identify meals that provide nutritional value while remaining within a user's specified spending limit.

### **Time-efficient meal planning**

Users can obtain a daily or weekly meal plan without manually searching for recipes.

### **Food utilization**

The system can recommend recipes based on ingredients already available to the user, potentially reducing food waste.

### **Application of AI**

The project demonstrates the practical integration of machine learning and recommendation techniques into a real-world problem.

## **1.4 Scope**

The scope of NutriSmart AI includes:

- User registration and profile creation.
- Dietary preference selection.
- Nutrition goal selection.
- Basic calorie and nutrient estimation.
- Recipe recommendation.
- Ingredient-based recipe search.
- Allergy and restriction filtering.

- Budget-based meal planning.
- Daily and weekly meal plans.
- Nutritional summaries.
- Estimated meal costs.
- Dashboard visualization.
- Recipe ranking and recommendation.

## **Out of Scope**

The project does not replace a qualified dietitian or doctor. It does not diagnose diseases or prescribe medical diets. Nutritional calculations are intended for general planning and educational purposes.

The accuracy of cost estimation may also depend on the food-price dataset used by the system and the geographical location of the user.

## **1.5 Methodology Overview**

The proposed methodology consists of several stages.

### **Step 1 – User Input**

The user provides information such as age, gender, activity level, dietary preference, nutritional goal, allergies, available ingredients, and budget.

### **Step 2 – User Profile Creation**

The system creates a personalized nutrition profile.

### **Step 3 – Nutrition Estimation**

The system estimates approximate energy and nutritional requirements using established calculation methods.

#### **Step 4 – Recipe Filtering**

Recipes that violate allergies, dietary preferences, or restrictions are removed.

#### **Step 5 – AI Recommendation**

Suitable recipes are ranked according to nutritional compatibility, user preferences, ingredients, and previous interactions.

#### **Step 6 – Budget Optimization**

The system attempts to construct meals that remain within the user's specified budget.

#### **Step 7 – Meal Plan Generation**

Daily or weekly meal schedules are generated.

#### **Step 8 – Feedback**

The user can rate, save, reject, or modify recommendations. This feedback can be used to improve future recommendations.

# Chapter 2: Problem Identification and Analysis

## 2.1 Description of the Problem

Modern users face several challenges when attempting to maintain healthy eating habits.

The first challenge is lack of personalization. Generic diet plans do not necessarily match individual nutritional requirements or preferences. The second challenge is recipe discovery. Users may know that they want healthy food but may not know what recipes to prepare. The third challenge is budget limitation. A meal can be nutritionally suitable but financially impractical. The fourth challenge is ingredient availability. Users often have ingredients at home but are unable to identify suitable recipes. The fifth challenge is dietary restrictions. Vegetarians, vegans, individuals with allergies, and users with specific food preferences require appropriate filtering.

The sixth challenge is manual planning. Calculating nutritional values and creating weekly meal schedules manually requires considerable time. Existing solutions often address individual aspects such as calorie tracking or recipe discovery, but an integrated solution combining nutrition + recipes + budget + personalization can provide greater practical value.

## 2.2 Evidence of the Problem

Poor dietary habits and nutrition-related problems are significant global concerns. The World Health Organization emphasizes the importance of a healthy diet and recommends appropriate consumption of fruits, vegetables, whole grains, legumes, nuts, and other nutritious foods.

At the same time, food affordability influences people's food choices. Users with limited budgets may prioritize low-cost food over nutritional quality.

Digital nutrition applications have increased access to calorie and recipe information, but many systems still depend heavily on manually entered information or provide recommendations that are not sufficiently personalized.

## Existing limitations

| Existing Approach       | Limitation                                            |
|-------------------------|-------------------------------------------------------|
| Traditional diet plans  | Not highly personalized                               |
| Not highly personalized | Focus primarily on tracking                           |
| Generic AI chatbots     | Recommendations may not be systematically constrained |
| Manual meal planning    | Time-consuming                                        |
| Shopping lists          | May not optimize nutritional value and cost together  |

## 2.3 Stakeholders

| Stakeholder                | Interest/Requirement                         |
|----------------------------|----------------------------------------------|
| Students                   | Affordable and easy meal planning            |
| Working Professionals      | Quick and convenient nutrition planning      |
| Families                   | Budget-conscious meal planning               |
| Fitness Users              | Nutrition-oriented meal recommendations      |
| Home Cooks                 | Recipe discovery using available ingredients |
| Nutrition Professionals    | Potential decision-support tool              |
| Food/Nutrition Researchers | AI-based nutrition analysis                  |
| Developers                 | AI and recommendation-system implementation  |

## 2.4 Supporting Data/Research

The system can be supported using recognized nutritional databases and research literature.

Potential data sources include:

- USDA FoodData Central.
- Indian food composition databases where available.
- Public recipe datasets.
- Nutritional research publications.
- Food-price datasets.
- WHO healthy-diet guidance.

The project can use these datasets to create a structured food database containing attributes such as:

- Food name
- Calories
- Protein
- Carbohydrates
- Fat
- Fiber
- Vitamins/minerals where available
- Ingredients
- Dietary category
- Preparation time
- Estimated cost

The use of structured nutritional data improves the consistency of recommendations.

# Chapter 3: Solution Design and Implementation

## 3.1 Development and Design Process

The project follows an iterative software-development approach.

### **Phase 1 – Requirement Analysis**

The requirements of users and system functions are identified.

### **Phase 2 – Dataset Collection**

Food, nutritional, recipe, and price information is collected.

### **Phase 3 – Data Preprocessing**

The collected data is cleaned, normalized, and transformed into a consistent format.

### **Phase 4 – System Design**

The architecture, database structure, recommendation pipeline, and user interface are designed.

### **Phase 5 – AI Development**

Recommendation and optimization algorithms are implemented.

### **Phase 6 – Front-End Development**

A user-friendly interface is developed.

### **Phase 7 – Integration**

The frontend, backend, database, recommendation engine, and budget module are integrated.

### **Phase 8 – Testing**

Functional, usability, performance, and recommendation tests are performed.

### **Phase 9 – Evaluation**

The system's outputs are evaluated using predefined parameters.

## 3.2 Tools and Technologies Used

Python is particularly suitable because of its extensive support for machine learning, data processing, and recommendation-system development.

| Category             | Technology                    |
|----------------------|-------------------------------|
| Programming Language | Python                        |
| Frontend             | HTML, CSS, JavaScript / React |
| Backend              | Flask / FastAPI               |
| Database             | MySQL / SQLite                |
| AI/ML                | Scikit-learn                  |
| Data Processing      | Pandas, NumPy                 |
| Visualization        | Matplotlib / Chart.js         |
| Development          | VS Code / Jupyter Notebook    |
| Version Control      | Git/GitHub                    |
| Dataset              | Food and recipe datasets      |
| Deployment           | Cloud platform/local server   |



### **3.3 Solution Overview**

NutriSmart AI consists of five major components:

#### **1. User Profile Module**

Collects:

- Age
- Gender
- Height
- Weight
- Activity level
- Dietary preference
- Food allergies
- Nutritional goal
- Budget

#### **2. Nutrition Analysis Module**

The system estimates approximate calorie requirements and creates nutritional targets.

#### **3. Recipe Recommendation Module**

The system identifies recipes matching user preferences and nutrition requirements.

#### **4. Budget Meal Planner**

The system evaluates estimated ingredient costs and generates meals within the user's budget.

#### **5. Dashboard**

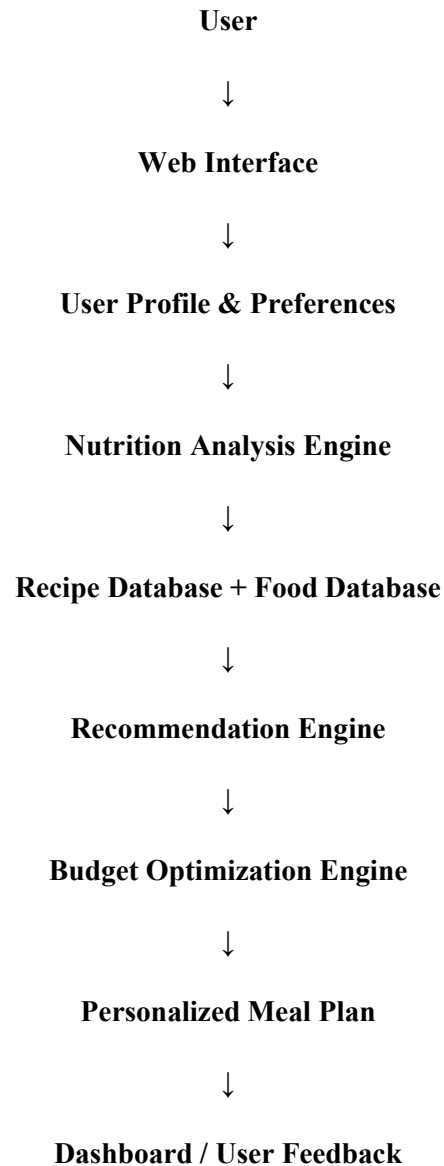
The dashboard displays:

- Daily calorie target
- Nutritional breakdown
- Recommended recipes

- Meal schedule
- Estimated cost
- Weekly budget usage
- Progress information

### 3.4 System Architecture

The system architecture follows this flow:



The architecture separates the presentation, application, data, and AI layers, making the system easier to maintain and extend.

## **3.5 Functional Modules**

### **Module 1: User Authentication**

Users can create accounts and securely access their personalized profiles.

### **Module 2: Nutrition Profile**

The system stores relevant user information and nutritional goals.

### **Module 3: Nutrition Calculator**

The module estimates approximate daily energy requirements.

### **Module 4: Recipe Recommendation**

Recipes are ranked according to user requirements.

### **Module 5: Ingredient-Based Recommendation**

Users can enter ingredients they already have, and the system recommends possible recipes.

### **Module 6: Budget Planner**

The system considers the maximum amount a user wants to spend.

### **Module 7: Meal Scheduler**

Recommended meals are organized into breakfast, lunch, dinner, and optional snacks.

### **Module 8: Dashboard**

The dashboard provides a visual summary of nutrition and budget information.

### **Module 9: Feedback System**

Users can provide ratings or indicate whether recommendations were useful.

### 3.6 AI and Recommendation Methodology

The intelligence of NutriSmart AI can be implemented using a combination of **content-based recommendation, similarity analysis, filtering, and optimization**.

For example, each recipe can be represented using features such as:

- Calories
- Protein
- Carbohydrates
- Fat
- Fiber
- Ingredients
- Cuisine
- Dietary type
- Preparation time
- Cost

A similarity-based recommendation approach can compare the user's nutritional profile with recipe characteristics.

A recommendation score can conceptually be represented as:

**Recommendation Score = Nutritional Match + Preference Match + Ingredient Match + Budget Match – Restriction Penalty**

Recipes containing allergens or prohibited ingredients receive a strong penalty or are removed completely.

#### **Budget Optimization**

Suppose a user provides a weekly budget of ₹1,500. The system attempts to select meals whose total estimated cost does not exceed the budget while maintaining reasonable nutritional coverage.

The optimization objective can be represented as:

## Minimize Cost

subject to:

- Nutritional requirements being approximately satisfied.
- Dietary restrictions being respected.
- Daily calorie targets being considered.
- User preferences being respected.
- Total budget remaining within the specified limit.

This transforms meal planning from a simple recipe search into a constrained recommendation problem.

## 3.7 Engineering Standards Applied

Several engineering and software standards can be considered in the development of NutriSmart AI.

| Standard                 | Application                           |
|--------------------------|---------------------------------------|
| ISO/IEC 25010            | Software quality evaluation           |
| ISO/IEC 27001 principles | Information security considerations   |
| IEEE 29148               | Requirements engineering              |
| IEEE 1016                | Software design description           |
| IEEE 829 concepts        | Software testing documentation        |
| OWASP principles         | Web application security              |
| APA                      | Academic references and documentation |

## **ISO/IEC 25010**

The system can be evaluated based on:

- Functional suitability
- Performance efficiency
- Compatibility
- Usability
- Reliability
- Security
- Maintainability
- Portability

## **IEEE 29148**

Requirements can be documented systematically, including functional and non-functional requirements.

## **OWASP**

Security practices can be incorporated to reduce risks such as unauthorized access, insecure authentication, and improper input handling.

## **3.8 Solution Justification**

The proposed solution is justified because it addresses the problem from multiple dimensions.

A nutrition-only application may tell users how many calories they need but may not tell them what affordable meal to prepare.

A recipe application may provide thousands of recipes but may not consider individual nutritional requirements.

A budget application may calculate food expenditure but may not evaluate nutritional suitability.

NutriSmart AI integrates these components.

The combination provides a more practical decision-support system:

**Personal Profile → Nutrition Target → Recipe Recommendation → Cost Analysis → Meal Plan**

This makes the system more useful for everyday meal decisions.

# Chapter 4: Results and Recommendations

## 4.1 Evaluation of Results

The effectiveness of NutriSmart AI can be evaluated using several parameters.

### Functional Accuracy

The system should correctly process user information and generate appropriate outputs.

### Recommendation Relevance

Recommended recipes should match:

- Dietary preference
- Nutritional requirements
- Allergies
- Budget
- Available ingredients

### Budget Compliance

Generated meal plans should remain within the user's specified budget.

### Usability

Users should be able to create profiles, obtain recommendations, and understand meal-plan information easily.

### Performance

The system should generate recommendations within an acceptable response time.

## 4.2 Output Parameters

| Parameter             | Expected Outcome                 |
|-----------------------|----------------------------------|
| User Profile Accuracy | Correct profile generation       |
| Calorie Estimation    | Personalized approximate target  |
| Recipe Relevance      | High matching recommendations    |
| Allergy Filtering     | Restricted food exclusion        |
| Budget Compliance     | Meal plan within specified limit |
| Meal Diversity        | Multiple suitable meal options   |
| Recommendation Speed  | Fast response                    |
| User Satisfaction     | Positive feedback                |
| Nutritional Coverage  | Balanced meal selection          |

For an academic prototype, evaluation can include metrics such as **Precision@K**, **Recall@K**, **recommendation relevance**, **budget compliance percentage**, and **user satisfaction score**.

For example:

$$\text{Budget Compliance (\%)} = \text{Plans Within Budget} / \text{Total Plans} \times 100$$

This provides a measurable way of evaluating the budget-planning component.



## **4.3 Challenges Encountered**

### **Challenge 1 – Nutritional Data Quality**

Food datasets may contain missing or inconsistent nutritional values.

**Solution:** Data cleaning and normalization techniques were applied.

### **Challenge 2 – Recipe Diversity**

Recipes may have different serving sizes and measurement units.

**Solution:** Standardized serving sizes and ingredient representations were used.

### **Challenge 3 – Budget Variation**

Food prices differ according to location, season, brand, and market.

**Solution:** Cost estimation is treated as an approximate value rather than a fixed price.

### **Challenge 4 – Recommendation Accuracy**

A recipe can be nutritionally appropriate but unsuitable for a user's taste.

**Solution:** User preferences and feedback are included in the recommendation process.

### **Challenge 5 – Allergy Handling**

Incorrectly recommending an allergenic ingredient can create serious problems.

**Solution:** Allergy restrictions are applied as hard filters before recommendation ranking.

## **4.4 Possible Improvements**

Future versions can include:

1. Real-time grocery prices.

2. Integration with online grocery platforms.
3. Mobile application support.
4. Voice-based meal recommendations.
5. Computer vision for food recognition.
6. Barcode-based food tracking.
7. More advanced machine-learning models.
8. Reinforcement-learning-based personalization.
9. Regional Indian cuisine support.
10. Seasonal ingredient recommendations.
11. Automatic grocery-list generation.
12. Food-waste prediction.
13. Personalized meal preparation time optimization.
14. Multilingual support.
15. Integration with wearable devices.

## **4.5 Recommendations**

The following recommendations are proposed for future deployment:

- Use verified nutritional databases.
- Regularly update food-price information.
- Provide transparent explanations for recommendations.
- Clearly communicate that the system is not a medical diagnosis tool.
- Implement strong authentication and data protection.
- Perform extensive usability testing.
- Include diverse regional food choices.
- Continuously improve recommendations using user feedback.
- Conduct larger-scale evaluation with real users.

# Chapter 5: Reflection on Learning and Personal Development

## 5.1 Key Learning Outcomes

### Academic Knowledge

The project provided an opportunity to apply concepts from computer science, artificial intelligence, database management, software engineering, data analysis, and web development.

The project demonstrated how theoretical concepts can be transformed into a practical application. Concepts such as machine learning, recommendation systems, optimization, database design, and software architecture were applied to solve a real-world problem.

### Technical Skills

During the development of NutriSmart AI, important technical skills were developed in:

- Python programming
- Data preprocessing
- Pandas and NumPy
- Machine learning
- Recommendation systems
- Database management
- API development
- Front-end development
- Data visualization
- Software testing
- Git/GitHub

Working with food and recipe datasets also improved understanding of data cleaning and feature engineering.

### Problem-Solving and Critical Thinking

The project required solving several interconnected problems rather than implementing a single algorithm.

For example, a recipe might satisfy nutritional requirements but exceed the user's budget. Another recipe might be affordable but violate dietary restrictions.

Therefore, the project encouraged a constraint-based approach to problem solving.

The development process improved the ability to break complex problems into smaller modules and evaluate different technical approaches.

## **5.2 Challenges Encountered and Overcome**

One of the major challenges was determining how different requirements should interact.

For example, prioritizing only calories could result in recommendations that users do not prefer. Prioritizing only budget could produce nutritionally weak recommendations.

This challenge was addressed by developing a multi-factor recommendation approach.

Another challenge was working with imperfect datasets. Missing values, inconsistent ingredient names, and different serving sizes required preprocessing.

Technical difficulties during integration also helped improve debugging and testing skills.

These challenges strengthened persistence, analytical thinking, and the ability to systematically identify the cause of technical problems.

## **Collaboration and Communication**

If the project is developed as a team, collaboration becomes an important component.

Different members can be responsible for:

- Frontend development
- Backend development
- Dataset preparation
- AI model development
- Database management

- Testing
- Documentation

Effective communication is necessary to integrate these components successfully.

Team discussions also provide experience in explaining technical concepts, accepting feedback, managing responsibilities, and resolving disagreements.

## **5.3 Application of Engineering Standards**

Engineering standards helped establish a structured approach to software development.

Requirements engineering principles helped identify what the system should accomplish.

Software-quality principles helped evaluate usability, reliability, maintainability, performance, and security.

Security practices helped identify the importance of protecting user information.

Following structured engineering practices also improved documentation and testing.

The project demonstrated that engineering standards are not simply documentation requirements; they can directly influence system quality and reliability.

## **5.4 Insights into the Industry**

NutriSmart AI provided insight into how modern software products are developed.

Real-world applications require more than programming. They require:

- Requirement analysis
- User experience design
- Data management
- Security
- Testing
- Deployment
- Documentation

- Continuous improvement

The project also demonstrated the importance of responsible AI.

Since nutrition recommendations can influence user decisions, AI outputs should be transparent, carefully validated, and presented as general guidance rather than medical diagnosis.

The project increased understanding of the role of AI and data-driven systems in consumer applications and healthcare-adjacent technologies.

## 5.5 Conclusion of Personal Development

The NutriSmart AI project contributed significantly to both technical and personal development.

Technically, it improved knowledge of Python, AI, machine learning, databases, recommendation systems, web development, and data analysis.

Personally, the project improved time management, critical thinking, communication, problem solving, and persistence.

The project also helped develop a stronger understanding of how academic concepts can be applied to real-world problems.

The experience provides a foundation for future work in **Artificial Intelligence, Machine Learning, Data Science, Software Development, and intelligent application development.**

# Chapter 6: Conclusion

NutriSmart AI was proposed as an intelligent solution for personalized nutrition, recipe recommendation, and budget meal planning. The project addresses the limitations of conventional meal planning by integrating user-specific nutritional requirements, dietary preferences, recipe information, ingredient availability, and financial constraints into a unified platform.

The system uses a combination of nutrition analysis, recommendation techniques, filtering, and budget optimization to generate suitable meal recommendations. Instead of simply presenting a collection of recipes, NutriSmart AI attempts to determine which meals are most appropriate for a particular user.

The project's key value lies in its integrated approach. Personalization improves relevance, recipe recommendation improves convenience, and budget optimization improves affordability. The system can therefore assist users in making more informed everyday food choices.

The project also demonstrates the practical application of artificial intelligence and software engineering principles to a real-world problem. Through appropriate data management, recommendation algorithms, quality evaluation, security practices, and user-centered design, NutriSmart AI can be developed into a scalable intelligent nutrition platform.

Future versions could incorporate real-time food prices, wearable-device data, computer vision, multilingual support, regional cuisine databases, grocery integration, and more advanced AI models.

Overall, NutriSmart AI demonstrates the potential of intelligent computing to support healthier, personalized, and economically practical meal planning.

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# Appendices

## Appendix A – Sample User Inputs

| Input                | Example           |
|----------------------|-------------------|
| Age                  | 21                |
| Gender               | Female            |
| Height               | 160 cm            |
| Weight               | 55 kg             |
| Activity             | Moderate          |
| Goal                 | Maintain Weight   |
| Diet                 | Vegetarian        |
| Allergy              | Peanuts           |
| Daily Budget         | ₹250              |
| Available Ingredient | Rice, Tomato, Dal |

## Appendix B – Sample System Output

### Personalized Nutrition

**Estimated Daily Energy Requirement:** ~2,000 kcal

**Suggested Daily Distribution:**

- Breakfast: ~450 kcal
- Lunch: ~650 kcal
- Snack: ~250 kcal
- Dinner: ~650 kcal

## Recommended Meals

| Meal      | Recommendation             | Estimated Cost |  |
|-----------|----------------------------|----------------|--|
| Breakfast | Vegetable Oats             | ₹45            |  |
| Lunch     | Dal Rice + Vegetable Curry | ₹70            |  |
| Snack     | Banana + Curd              | ₹30            |  |
| Dinner    | Vegetable Chapati + Dal    | ₹75            |  |

**Estimated Daily Cost: ₹220**

**Budget: ₹250/day**

**Remaining Budget: ₹30**

## Appendix C – Suggested System Screens

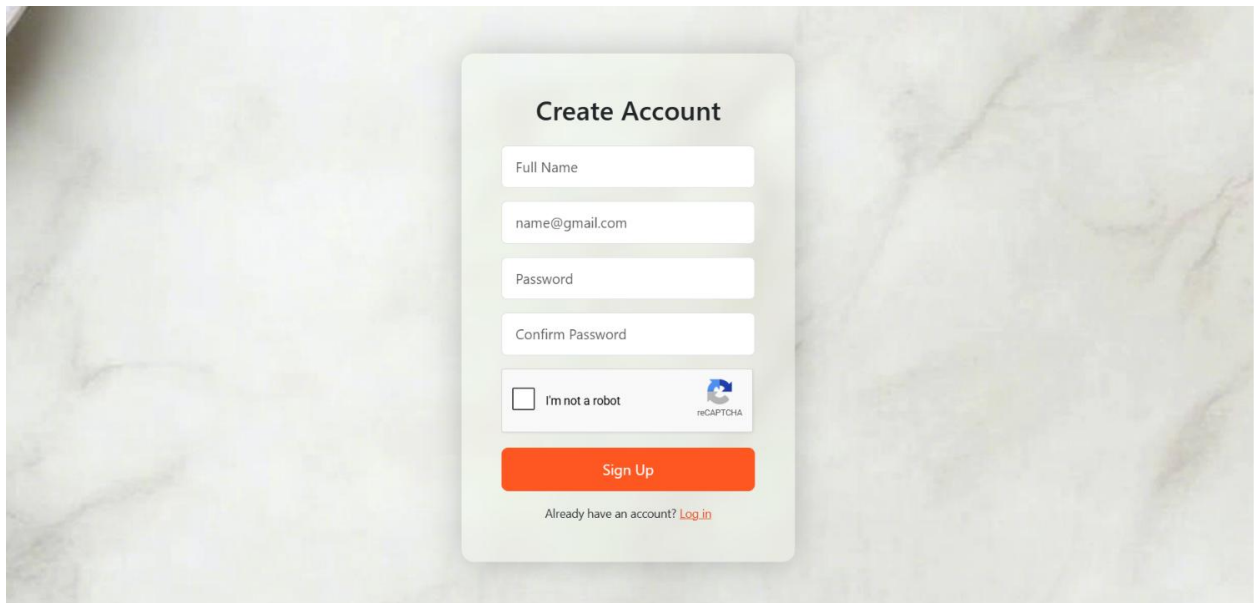
Your final implementation can include these screenshots:

1. Login/Register page
2. User profile page
3. Nutrition goal page
4. AI recommendation dashboard
5. Recipe recommendation page
6. Recipe details page
7. Budget meal planner
8. Weekly meal plan
9. Nutritional chart
10. Grocery list
11. User feedback page

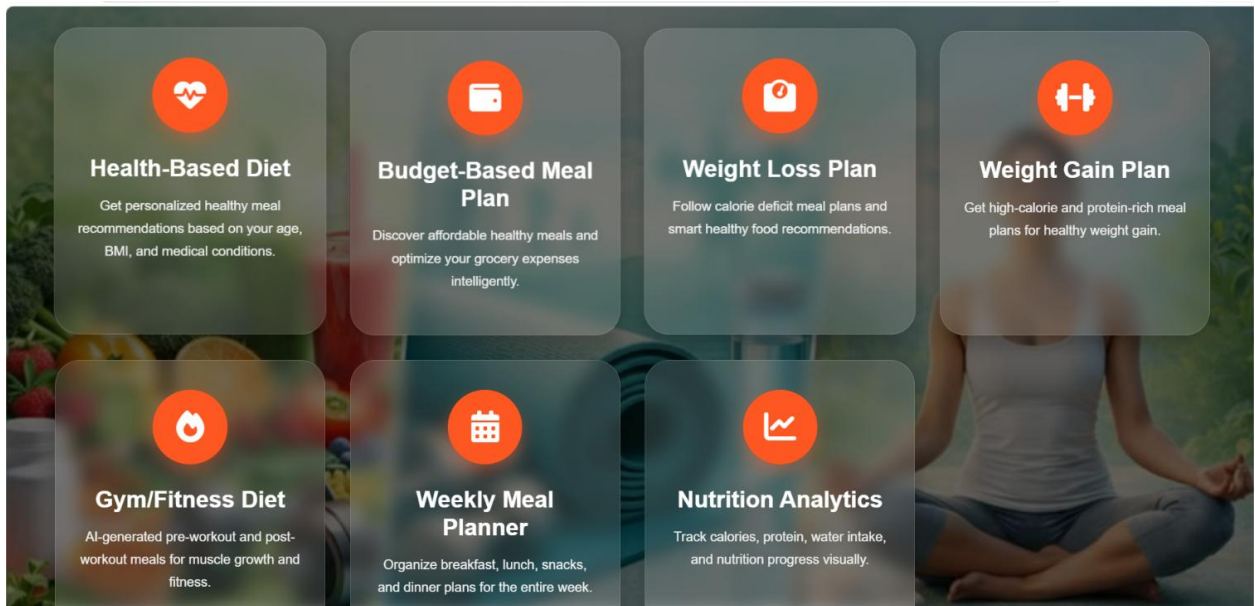
# Implementation

The screenshot shows a VS Code editor with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project structure with folders like 'backend', 'frontend', 'images', and 'recipes\_images'. The code editor displays the 'login.html' file, which contains HTML code for a login form and a 'Forgot Password Modal'. The login form includes a 'Log In' button and a link to 'Sign up'. The modal is a 'forgotPasswordModal' with a 'Reset Password' button and an email input field. The terminal shows the output of a 'uvicorn' command, indicating that the application is running on '127.0.0.1:59851'.

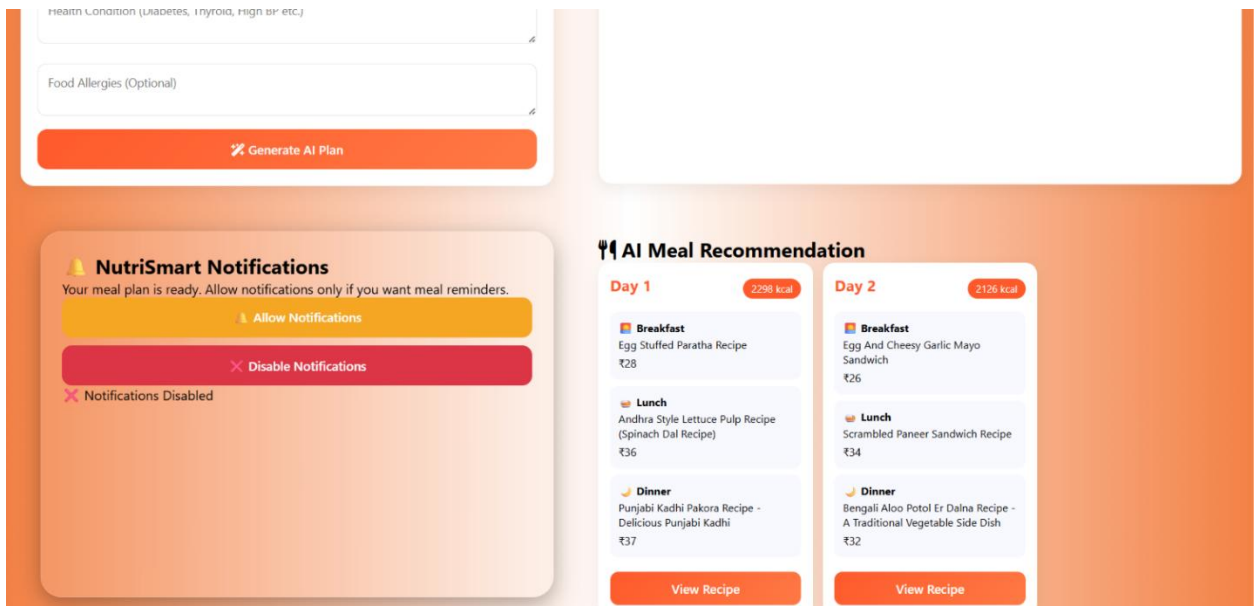
```
2 <html lang="en">
96 <body>
105 <div class="d-flex justify-content-center align-items-center min-vh-100">
106 <div class="glass-form">
108 <form id="loginForm" novalidate>
117
118 <button type="submit" class="btn btn-primary w-100">Log In</button>
119 <p class="mt-3 small">Don't have an account? <a href="/" style="color: #e64a19;">Sign up</a>
120 </p>
121 </div>
122 </div>
123 </div>
124
125 <!-- Forgot Password Modal -->
126 <div class="modal fade" id="forgotPasswordModal" tabindex="-1" aria-hidden="true">
127 <div class="modal-dialog modal-dialog-centered">
128 <div class="modal-content">
129 <div class="modal-header">
130 <h5 class="modal-title">Reset Password</h5>
131 <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
132 </div>
133 <div class="modal-body">
134 <input type="email" id="resetEmail" class="form-control" placeholder="Enter your Gmail">
135 <div class="text-danger mt-1" id="resetEmailError"></div>
136 </div>
137 <div class="modal-footer">
```



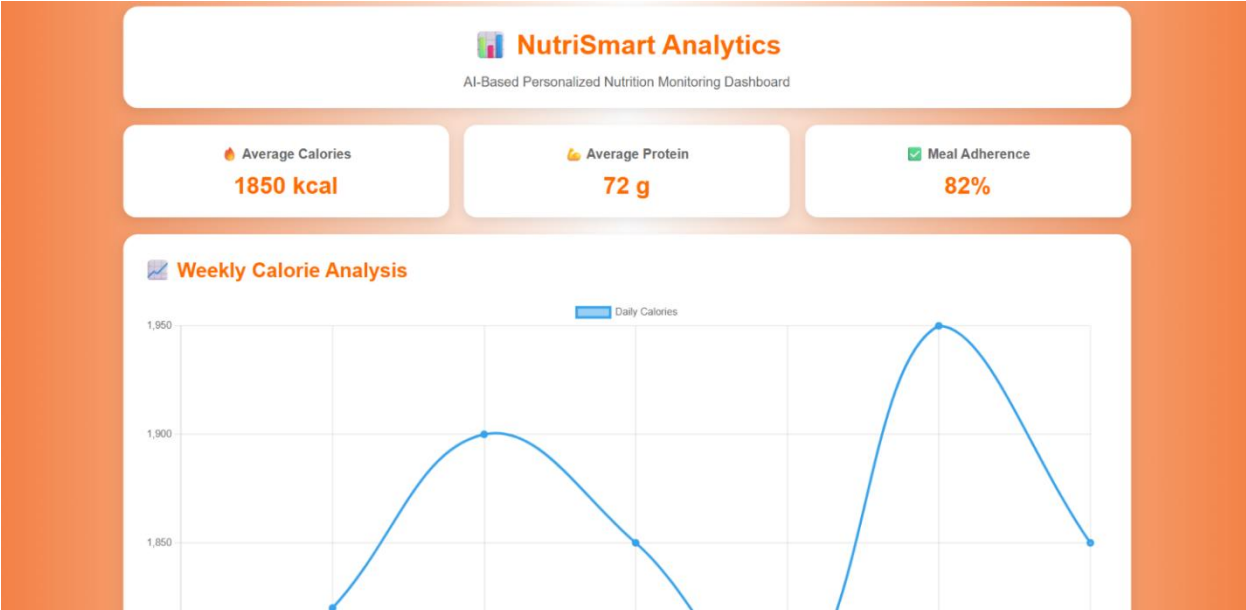
## Login Page



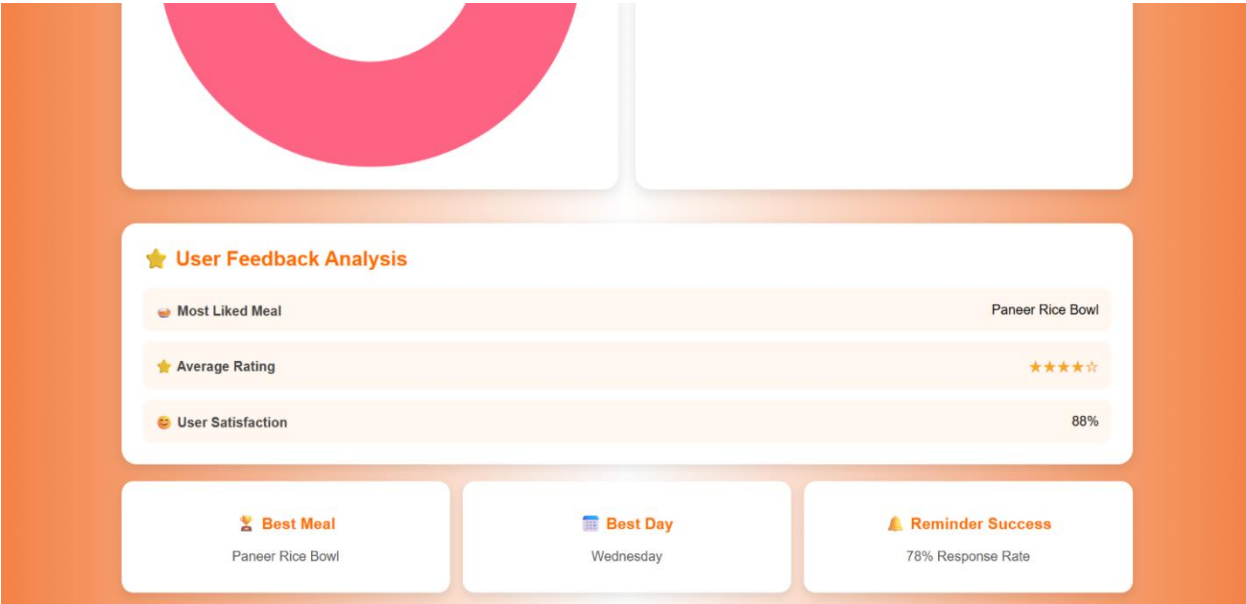
## Nutrition planings



## Budget recommendation



Analytics



Feedback